
Pivotal Tokens Are Linearly Predictable Before They Are Generated

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Abstract

Pivotal Token Search (PTS) identifies the individual tokens at which a language model’s probability of reaching a correct answer shifts sharply, but it does so offline and at a cost of tens of sampled rollouts per candidate position. We ask whether pivotality is recoverable from the model’s own residual stream *one position before* the token is emitted, which would turn an expensive offline analysis into a single dot product available during decoding. On Qwen3-0.6B over GSM8K, a linear probe read at $t-1$ attains **0.812 AUROC** (95% CI [0.739, 0.864], clustered by question) at predicting whether the next token is pivotal. The comparison that matters is against cheaper explanations: the probe exceeds a token-identity lookup table by 11 points and next-token entropy by 18, with a confidence interval whose lower bound lies above both. Pivotality is therefore not a restatement of the high-entropy “forking token” account of critical positions. We further find that the result is highly sensitive to regularization—AUROC moves monotonically from 0.698 to 0.812 as the inverse penalty C falls from 1 to 10^{-4} , with the probe direction rotating toward the contrastive mean-difference direction—and that no individual layer is special: under a null in which all layers are equally informative, the expected maximum over 29 layers *exceeds* the observed maximum. We release a rebuilt labelling pipeline that recovers $3.8\times$ more supervised positions from the same source data than the construction it replaces.

1 Introduction

A language model solving a multi-step problem does not distribute its risk evenly across the tokens it emits. Most continuations are near-determined by what came before; a few are decision points at which the eventual outcome genuinely changes. The Phi-4 technical report [Abdin et al., 2024] formalized this as *Pivotal Token Search* (PTS): a token t is pivotal when appending it to a prefix moves the estimated probability of eventually solving the task by more than a threshold,

$$|\Delta p| = |P(\text{success} \mid \text{prefix} + t) - P(\text{success} \mid \text{prefix})| > \tau. \quad (1)$$

Estimating either term requires sampling many completions and adjudicating each with an oracle, so PTS is an offline instrument. It labels data; it cannot watch a model as it decodes.

This paper asks whether that instrument can be amortized. Concretely: is pivotality *linearly decodable from the residual stream at position $t-1$* , before the token in question is produced? A probe read one step early is a fundamentally different object from one read at the pivotal token itself. The latter is a post-hoc classifier; the former is available while there is still a decision to influence, and costs one dot product against roughly 50×320 generated tokens for the oracle it approximates.

The question is only interesting if the answer is not already known for a cheaper reason. Two such reasons exist and we treat both as primary baselines rather than as controls to be mentioned in passing. *Token identity*: some tokens may simply be pivotal more often than others, in which case a lookup table suffices and no representation is implicated. *Next-token uncertainty*: high-entropy “forking”

tokens are the incumbent notion of a critical position [Wang et al., 2025], and a probe that merely rediscovers entropy adds nothing.

Our contributions:

- A rebuilt labelling pipeline for PTS events. PTS samples several rollouts per question and they diverge; collapsing a question to a single canonical sequence, as prior constructions do, discards every pivot on a diverging branch. Treating each maximal branch as its own sequence recovers $3.8\times$ more supervised positions from identical source data (Section 3.1).
- Evidence that pivotality is linearly decodable at $t - 1$ at 0.812 AUROC, and that this exceeds both token identity and every next-token uncertainty statistic we tested (Section 5.1).
- A characterization of two ways this measurement goes wrong: regularization strength moves AUROC by 11 points monotonically, and “best layer” selection is indistinguishable from noise (Sections 5.2 and 5.3).
- A first look at the *signed* question—whether the direction of the impending shift is separately decodable—which shows signal above entropy but is not yet separable from token identity (Section 5.4).

2 Related Work

Pivotal tokens. Abdin et al. [2024] introduce PTS to build preference pairs for Phi-4 post-training; Sharma [2025] released an open implementation and the event datasets we build on. Both use PTS as a data-generation step. To our knowledge the representational question—whether the property PTS measures is present in the model’s activations beforehand—has not been asked.

Sentence-scale reasoning importance. Thought Anchors [Bogdan et al., 2025] identifies sentences with outsized influence on a reasoning trace via counterfactual resampling, receiver heads, and attention suppression. We work at token scale, which is where the $t - 1$ prediction problem is well posed: a sentence boundary is not a decoding step.

Uncertainty as the incumbent account. Wang et al. [2025] show that roughly 20% of tokens carry high entropy, act as branch points, and account for most of the gain from reinforcement learning with verifiable rewards. This makes next-token uncertainty the hypothesis any claim about token criticality must beat, and Section 5.1 treats it accordingly. Note an asymmetry we return to in Section 5.4: entropy is sign-blind by construction, so it cannot in principle account for a probe that separates helpful from harmful pivots.

Probing and steering. Linear probes on intermediate representations are standard [Alain and Bengio, 2018, Marks and Tegmark, 2023], as is contrastive activation addition [Rimsky et al., 2024, Huang, 2024]. Conditional steering [Lee et al., 2024] gates an intervention on a learned condition; our setting differs in what the gate conditions on—not a semantic category of the input, but a prospective, outcome-grounded property of the very next token.

Value versus leverage. Process reward models and value probes [Li et al., 2024] estimate $P(\text{success} \mid \text{prefix})$, a *level*. Pivotality is $|\Delta p|$ at the next token, a *sensitivity*. The distinction is not rhetorical: a state can be highly favourable and completely insensitive, or balanced and about to tip.

3 Method

3.1 Labels

Each PTS event names a prefix and a token whose appending moved the success probability past τ . We take the position immediately *preceding* that token as a positive example, and sample non-pivotal positions from the same answer span as negatives at a 1:1 ratio.

Two construction details matter more than they appear to.

Branches, not a canonical sequence. PTS samples `max_generations` rollouts per question, and they diverge after a few tokens. A construction that collapses a question to its longest rollout and matches the others against it as string prefixes silently discards every pivot on a diverging branch. On the released Qwen3-0.6B events, 104 questions carry 1,245 pivotal tokens spread over 447 distinct branches, and the collapsing construction retains 318 of them. We instead treat every maximal token sequence as its own example, recovering 1,216 positions— $3.8\times$ more supervision from identical data. Because branches share prefixes, and the residual stream at position p depends only on tokens $0 \dots p$, we label each distinct context exactly once; otherwise shared early positions would enter training several times.

Indices are verified, not inferred. Released events predating the current schema carry no absolute position, so the index must be recovered by re-tokenizing the prefix. This is unsafe: byte-pair encoding does not guarantee $\text{enc}(a + b) = \text{enc}(a) + \text{enc}(b)$, and a pivot token such as “.” frequently merges with the preceding word. We verify each recovered index against the reported token id, and discard rows that fail. On the released data this rejects 1 row in 1,245 rather than mislabelling it, and every retained label was confirmed to sit immediately before a reported pivot.

3.2 Label noise in the oracle

$\widehat{\Delta p}$ is a difference of two binomial estimates from S rollouts each, accepted when $|\widehat{\Delta p}| \geq \tau$. Under the null of no true change, $\text{Var}(\widehat{\Delta p}) = 2p(1 - p)/S$, so at the reference setting $S = 50$, $\tau = 0.2$ the acceptance threshold sits at only 2σ : roughly a 4.6% false-acceptance rate per tested position at $p = 0.5$. With tens of positions tested per rollout, a non-trivial share of the positive class is sampling noise. This bounds what any probe can achieve and should be read alongside every number in Section 5. [TODO: quantify the clean-subset gap once we store sufficient statistics rather than verdicts.]

3.3 Probes and selection discipline

Probes are ℓ_2 -regularized logistic regressions on the residual stream, fit on standardized features and then converted back to raw activation space so that the resulting vector is usable as a direction. We also report the contrastive mean-difference direction $\mu_+ - \mu_-$, computed on training data.

Two disciplines are enforced throughout, because relaxing either inflates the headline substantially (Sections 5.2, 5.3). **Selection is confined to training data:** layer and C are chosen on an inner split of the training questions, and the test split is used once. **Confidence intervals cluster by question:** positions drawn from one reasoning trace are not independent, so intervals come from a bootstrap over questions rather than rows.

3.4 Baselines

Token identity, as a Laplace-smoothed log-odds lookup over the token at $t - 1$, and separately as one-hot logistic regression. Both are fit on train and scored on held-out rows; scored in-sample, such a table memorizes and reports spurious skill. *Next-token uncertainty*: entropy, negative top-2 margin, and negative top-1 probability, computed from the model’s own distribution at $t - 1$ and used directly as detectors in the direction the literature uses them. *Random direction*: the mean of eight random unit projections.

4 Experimental Setup

Model: Qwen3-0.6B [Qwen Team, 2025], 28 layers, $d = 1024$. Events: the released `codellion/Qwen3-0.6B-pts` token-granularity set over GSM8K [Cobbe et al., 2021], generated with raw-text conditioning and therefore without the model’s extended-thinking mode. Activations are read at all 29 residual-stream positions (`hidden_states[L]`, index 0 being the embedding output) in `bfloat16` at the labelled positions only.

After construction: 106 question groups over 425 branches, yielding 1,216 pivotal and 1,187 non-pivotal positions. Split by question into 1,963 training rows (85 questions) and 440 test rows (21

Table 1: Predicting whether the *next* token is pivotal, from the residual stream at $t - 1$. Qwen3-0.6B, GSM8K, 440 held-out positions from 21 unseen questions. Intervals are 95% bootstrap, clustered by question.

Detector	AUROC	95% CI	Accuracy
Linear probe ($L=18$, $C=10^{-4}$)	0.812	[0.739, 0.864]	0.752
Mean-difference direction	0.784	[0.718, 0.837]	0.745
Token identity (frequency table)	0.701	—	0.652
Token identity (one-hot LR)	0.697	—	0.652
Next-token entropy	0.628	—	0.509
Negative top-1 probability	0.617	—	0.491
Negative top-2 margin	0.609	—	0.491
Random direction	0.449	—	0.455

Table 2: Effect of the inverse penalty C at the selected layer. As C falls the probe direction rotates toward the mean-difference direction and performance improves monotonically.

C	AUROC	$\cos(w, \mu_+ - \mu_-)$	$\ w\ $
10^{-4}	0.812	0.639	0.07
10^{-3}	0.807	0.338	0.23
10^{-2}	0.773	0.158	0.89
10^{-1}	0.730	0.095	2.78
1	0.698	0.072	7.70

questions), the latter 50.9% positive. Extraction takes 7.3 minutes on an Apple M3 laptop and produces 143 MB.

5 Results

5.1 Pivotality is decodable at $t - 1$, and is not token identity

Table 1 reports every detector on the held-out split at the selected layer ($L = 18$) and penalty ($C = 10^{-4}$).

The probe exceeds token identity by 11 AUROC points and next-token entropy by 18, and the lower bound of its interval, 0.739, lies above both point estimates. Two readings follow. Pivotality is present in the residual stream one step before the token that realizes it, and it is not merely a property of which token is about to be emitted. And it is not the forking-token phenomenon: every uncertainty statistic we tested lands between 0.61 and 0.63, well below the probe.

5.2 Regularization moves the result by 11 points

Table 2 shows a monotone 11-point swing across four orders of magnitude, with cosine similarity to the mean-difference direction rising from 0.07 to 0.64 as the penalty tightens. A weakly regularized probe still classifies, but it does so along a direction nearly orthogonal to the contrastive one. This is consistent with the observation that probe accuracy and causal relevance can come apart, and it is a concrete reason to report the regularization path rather than a single operating point.

5.3 No layer is special

Per-layer test accuracy at the selected C spans 0.670–0.759 with mean 0.730; AUROC spans 0.709–0.821 with mean 0.792. To ask whether the best layer is meaningfully best, we simulate a null in which all 29 layers are equally informative, each measured on 440 rows. The expected maximum under that null is **0.772**, which *exceeds* the observed maximum of 0.759; the observed maximum is at the 93rd percentile from below, i.e. entirely unremarkable. The signal is distributed across the

mid-to-late stack rather than localized, and reporting a single “pivotal layer” would be an artifact of maximizing over correlated noise.

The size of that artifact is measurable. Choosing the layer on held-out data and reporting it there gives 0.821; choosing it properly on training data and reporting on test gives 0.812—a gap of 0.009. When we deliberately let selection leak, the same gap widens to 0.063. Both the discipline and its absence are visible in the numbers.

5.4 The signed question: promising, not established

Pivotality as defined in Eq. 1 is symmetric: a token that rescues an answer and one that ruins it receive the same label. The more useful question is whether the *sign* is decodable at $t - 1$ —whether the model represents, one step early, that it is about to make a mistake.

Restricting to pivotal positions and relabelling by $\text{sign}(\Delta p)$ gives 992 training and 224 test rows (45.2% helpful). A probe reaches 0.717 AUROC [0.630, 0.790], against 0.586 for entropy and 0.366 for a random direction. That entropy is near chance here is expected and is the point: entropy cannot distinguish a helpful branch point from a harmful one, so a signed result cannot be explained by it.

However, token identity reaches 0.687, which lies comfortably inside the probe’s interval. At 21 test questions this is equally consistent with a real effect and with none. We therefore report the signed probe as unresolved rather than as a finding. It is the clearest motivation for scaling the number of questions, which we take up next.

6 Limitations

Results are from a single 0.6B model on a single task family, with 21 held-out questions; intervals are correspondingly wide and the signed result is underpowered. The oracle’s own labels are noisy (Section 3.2) in a way we have quantified analytically but not yet empirically. The PTS acceptance filter retains only questions whose baseline success probability lies in a middle band, so pivot density is confounded with task difficulty relative to model capability, and the label distribution is not representative of all questions. Linear probes test linear decodability only. Finally, everything here is correlational: we have not yet shown that the direction the probe identifies is causally used by the model, and prior work cautions that probe accuracy does not imply causal relevance. **[TODO: Causal validation via probe-gated intervention is the next experiment.]**

7 Conclusion

Pivotality—an outcome-grounded property that has so far only been measured by expensive offline sampling—is linearly present in a model’s residual stream one position before the token that realizes it, at 0.812 AUROC, and is separable from both token identity and next-token uncertainty. The measurement is fragile in two specific ways we characterize: it depends strongly on regularization, and apparent layer specialization does not survive a null comparison. Whether the *direction* of the impending shift is also decodable remains open at this sample size, and is the natural target for scale.

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A Reproduction

All numbers in Section 5 come from `artifacts/probes_v2/qwen3-0.6b_report.json`, produced by `scripts/build_and_extract.py` followed by `scripts/evaluate_probes_v2.py`.
 [TODO: Add layer sweep figure and regularization-path figure.]